PROJECT REPORT ON

MEDRED

MEDICINE DONATION AND DISTRIBUTION PLATFORM

"Don't waste medicines, redistribute hope."

PROJECT REPORT & TECHNICAL GUIDE

Submitted in partial fulfillment of the requirements for the award of **Diploma in Computer Engineering.**

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CERTIFICATE

This is to certify that this project report entitled **"MEDRED: MEDICINE DONATION AND DISTRIBUTION PLATFORM"** is a bonafide record of the project work carried out by the students listed below, in partial fulfillment of the requirements for the award of Diploma in Computer Engineering, during the academic session 2024–2027.

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The work is submitted for academic review. The system is a web application prototype; its results should be treated as an informational aid and not as a substitute for professional medical or commercial advice.

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We are thankful to our Principal, **P. Srinivas, M.Tech**, for providing the opportunity and institutional support to undertake this work. We also thank the faculty members of the Department of Computer Engineering for their assistance and feedback.

We acknowledge the open-source technologies used in the MEDRED project, including PHP, PostgreSQL, Supabase, Bootstrap, Razorpay, Leaflet and Playwright. Finally, we thank our families and friends for their encouragement.

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ABSTRACT

MEDRED is a web-based medicine donation and distribution platform that connects verified medicine distributors and pharmacies with non-governmental organizations, medical shops, old-age homes, orphanages and charitable hospitals. A large quantity of usable medicine is wasted every year because distributors hold surplus or near-expiry stock while social organizations struggle to obtain medicines for the people they serve. MEDRED, whose guiding line is “Don't waste medicines, redistribute hope,” addresses this gap with a single, verified marketplace.

The application provides role-based access for administrators, distributors and organizations. Distributors publish stock with batch number, expiry date, price and quantity. Organizations search the store, add medicines to a cart, choose a delivery address, enter a delivery mobile number and pay through Razorpay (UPI, card or net banking) or cash on delivery. Every order passes through a controlled lifecycle: Pending Payment, Pending, Accepted, Packed, Shipped, Out for Delivery and Delivered. Stock is reserved when an order is placed and restored when it is cancelled or rejected. Expired and out-of-stock medicines are removed automatically and the distributor is notified.

The system is implemented with PHP 8 on Apache, PostgreSQL hosted on Supabase, Supabase Storage for uploaded documents, Bootstrap 5 and vanilla JavaScript, and is packaged as a Docker image for deployment on Render. It includes admin verification of documents with complete removal of rejected registrations, in-app notifications with an unread badge, reviews, reports with CSV export that exclude soft-deleted medicines, audit logs, persistent thirty-day sign-in, an SMTP email-OTP password-reset flow, address-first delivery navigation and performance optimizations for a hosted database. The platform was tested at the API, browser and live-deployment levels; all functional tests passed and cached read paths respond in milliseconds.

KEYWORDS

Medicine donation • Medicine redistribution • E-commerce • Order management • Razorpay • Supabase • PHP • PostgreSQL

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Executive Summary

MEDRED is a complete web application for redistributing usable medicines from verified distributors to verified social organizations. It replaces informal phone and spreadsheet coordination with a single platform that shows live stock, expiry dates, prices and order status.

The platform has three roles. Distributors manage their medicine inventory. Organizations browse the store, place orders and pay online or choose cash on delivery. Administrators verify accounts, review uploaded documents and monitor reports and audit logs.

The main outcomes of the project are:

- a verified network of distributors and recipient organizations;
- expiry-aware inventory with automatic removal of expired and out-of-stock medicines;
- an order lifecycle with stock reservation, cancellation, rejection and delivery tracking;
- server-verified Razorpay payments with UPI, card and net-banking support, plus cash on delivery;
- in-app notifications with an unread badge;
- reviews, reports with CSV export and a complete audit trail;
- reports that count only active medicines, so soft-deleted stock no longer inflates admin totals;
- a delivery mobile number saved with every order and reused for navigation;
- address-first navigation through Google Maps, with stored coordinates as a fallback;
- a thirty-day persistent sign-in with a signed, HttpOnly remember cookie;
- an SMTP email-OTP password-reset flow;
- rejection of pending registrations that removes the account, profile and uploaded document;
- Docker-based deployment on Render with Supabase PostgreSQL and Storage;
- caching and lazy database access that keep cached reads in the millisecond range.

The system was validated with API tests, browser tests, a role-based smoke test and live checks on the Render deployment. All functional checks passed.